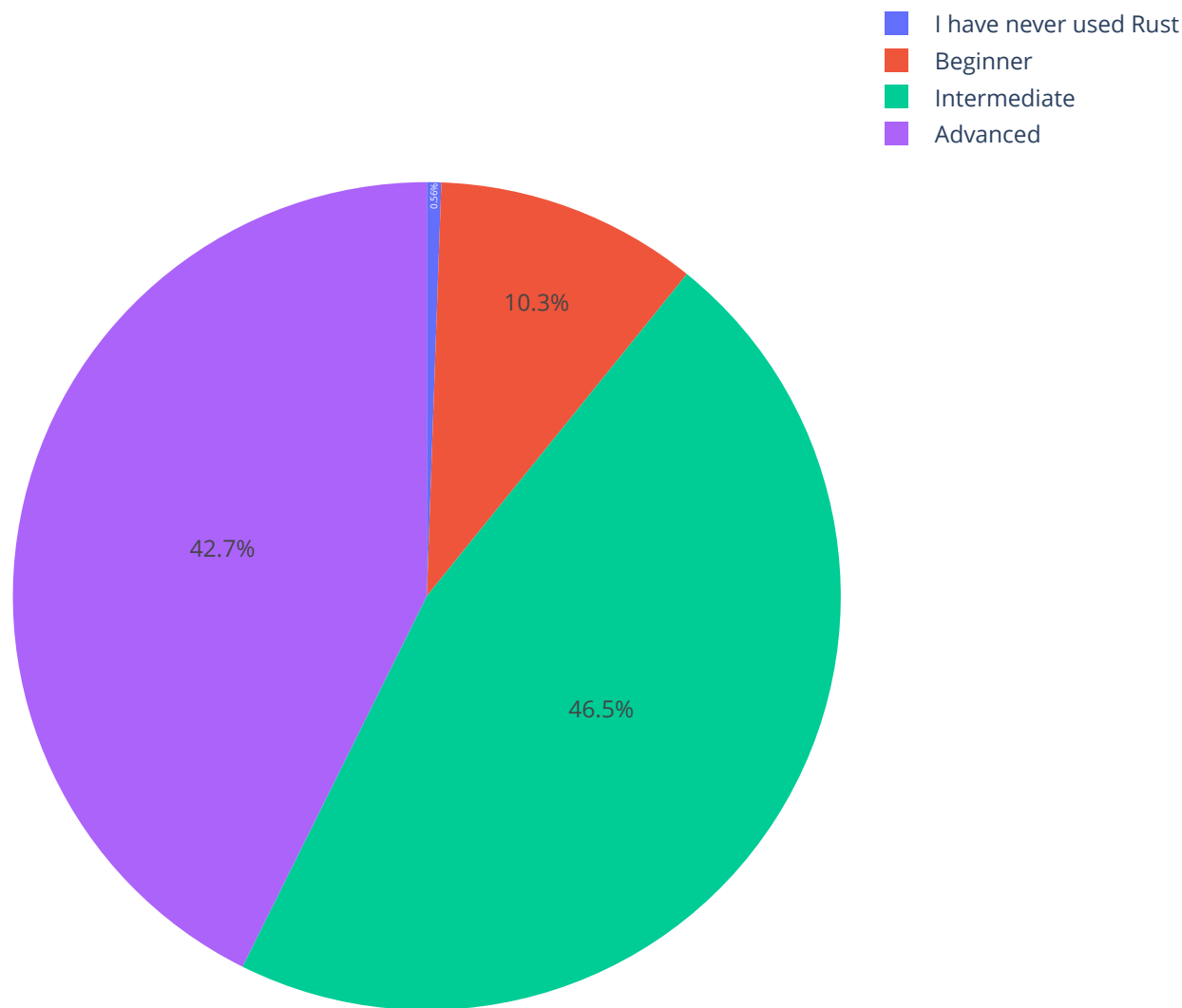


Rust Debugging survey 2026 report

How would you rate your Rust expertise?

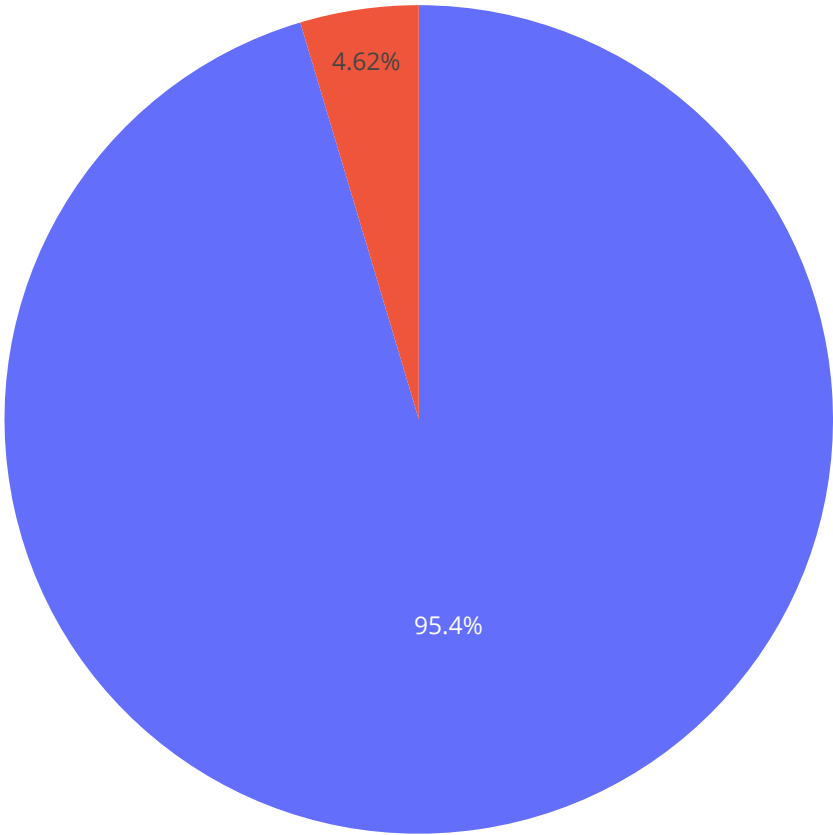
(total responses = 2321)



Do you currently use Rust?

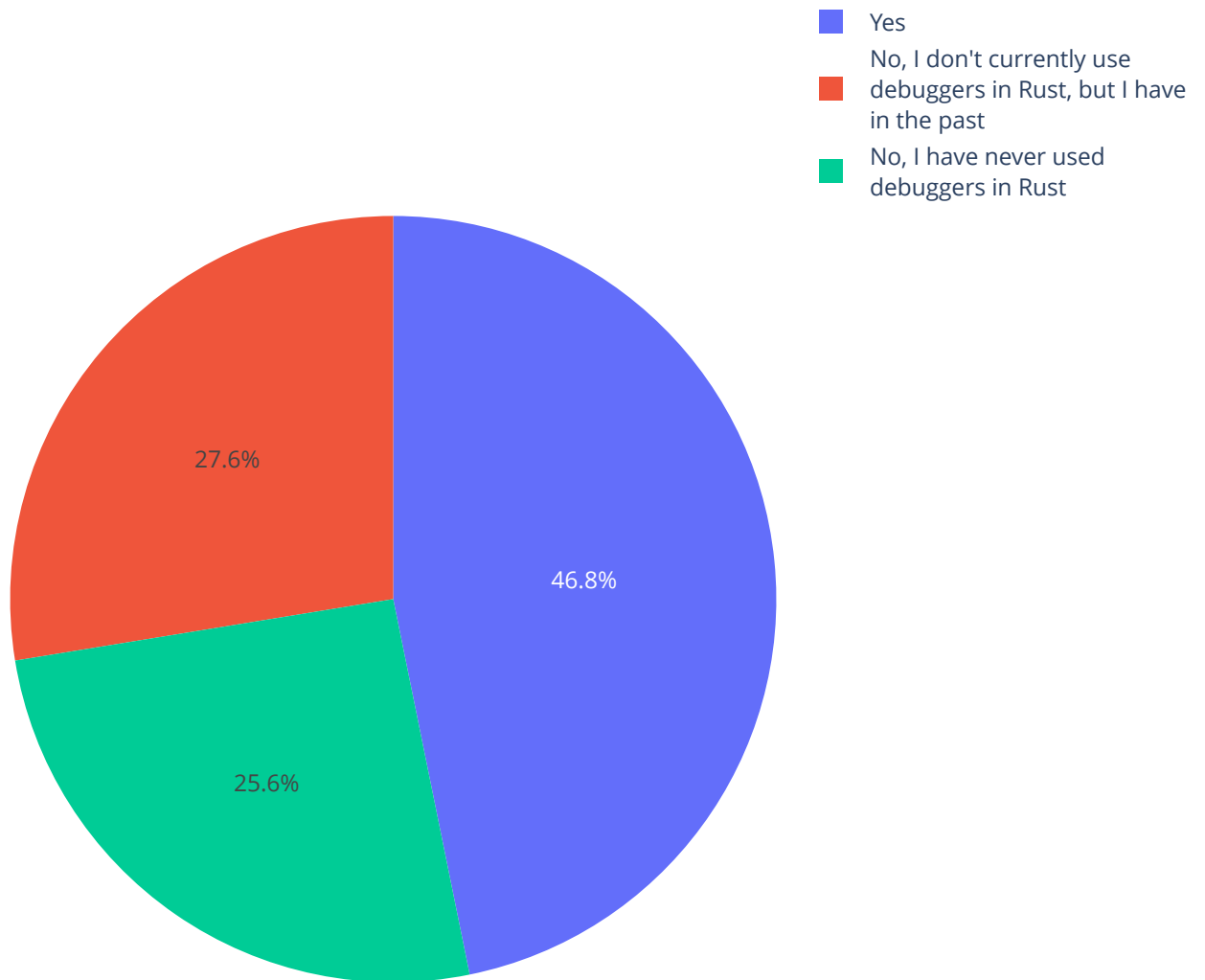
(total responses = 2295)

- Yes, I use Rust
- No, I don't currently use Rust, but I have in the past



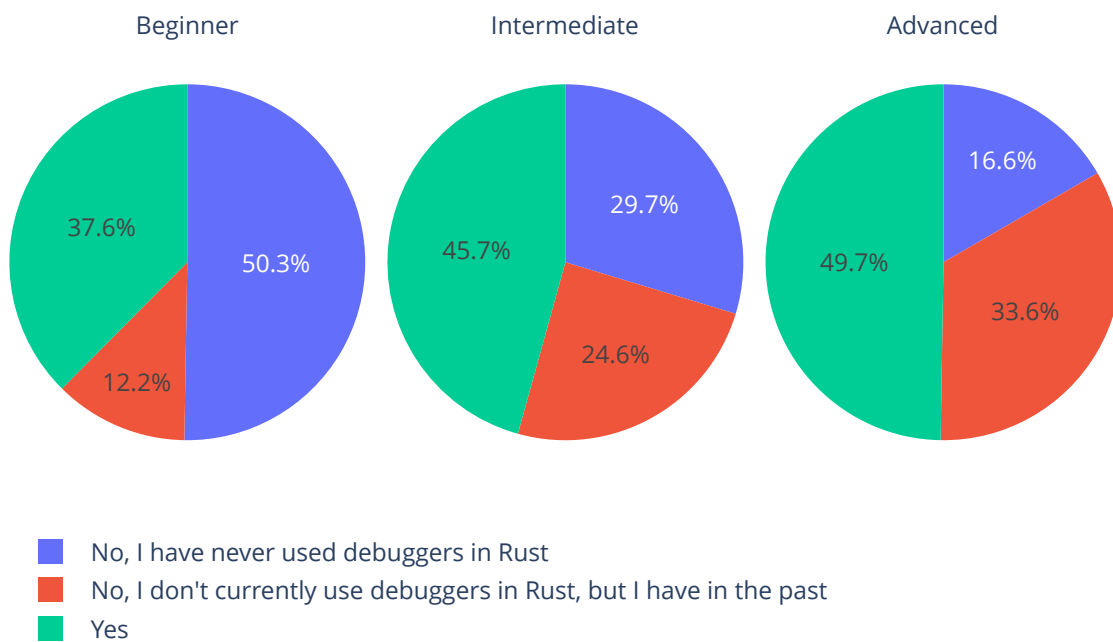
Do you use debuggers in Rust?

(total responses = 2173)



Do you use debuggers in Rust? (based on expertise)

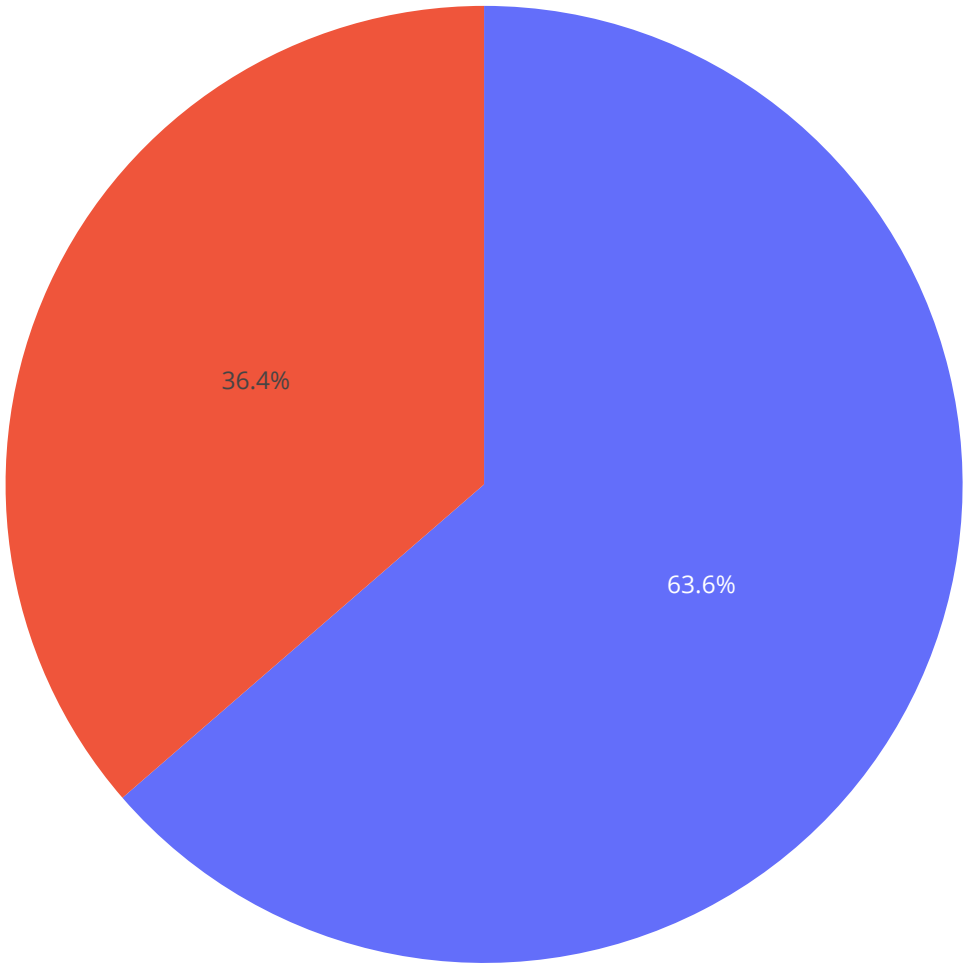
(total responses = 2173)



Did you use debuggers in Rust?

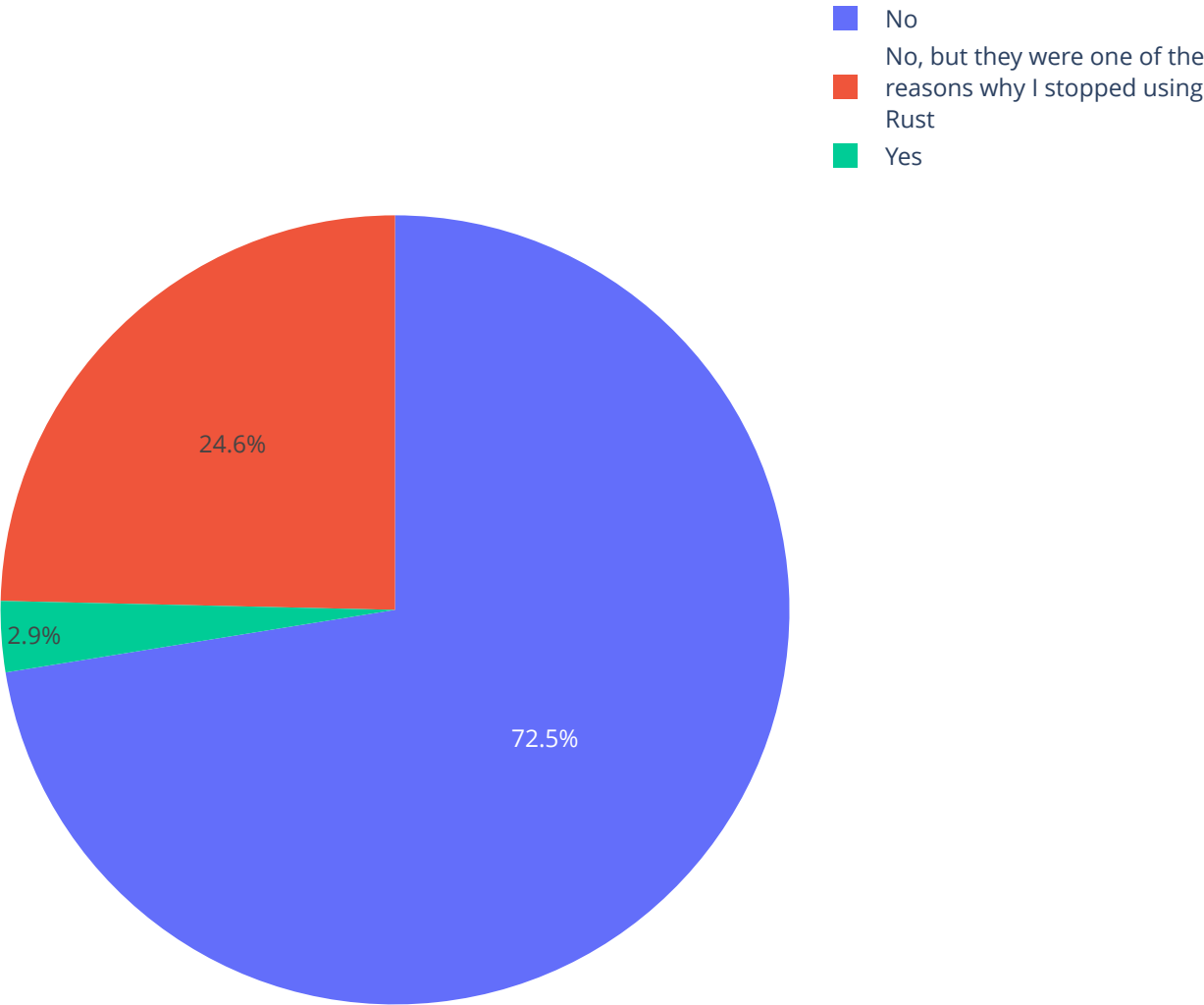
(total responses = 110)

- Yes, in the past
- No, never



Were issues with debugging support the primary reason why you stopped using Rust?

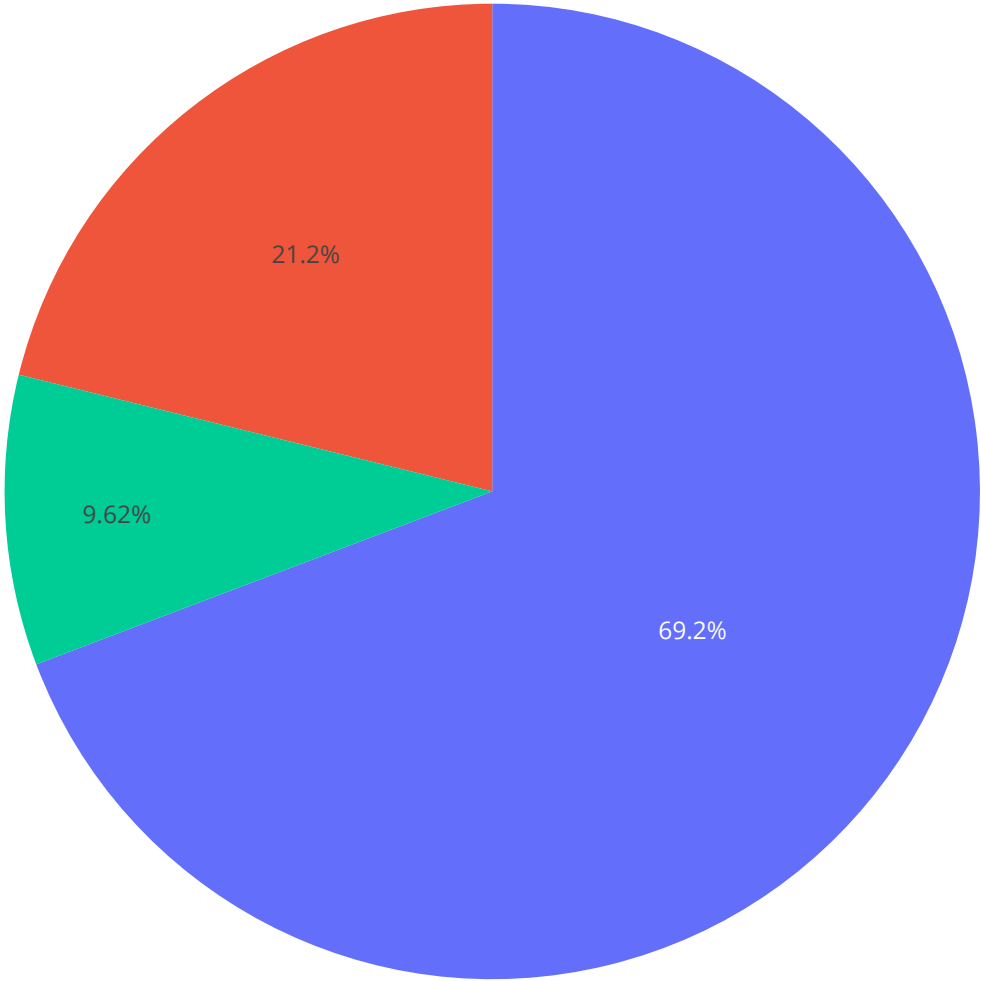
(total responses = 69)



Do you use debuggers in other programming languages?

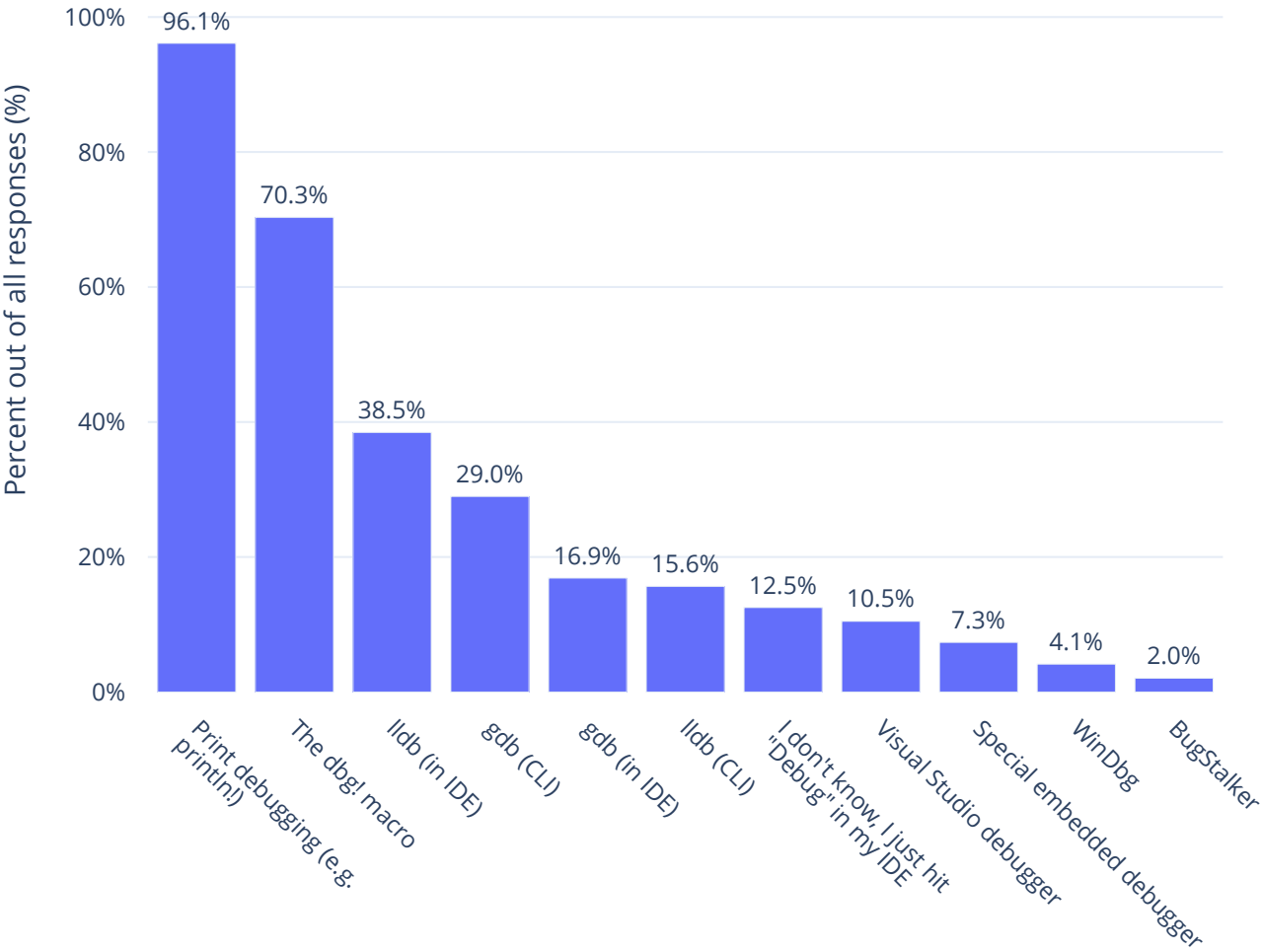
(total responses = 52)

- Yes
- Not anymore
- No, never



What tools and workflows do you use to debug Rust programs?

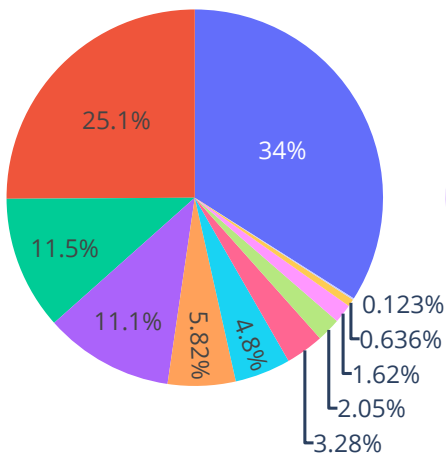
(total responses = 2109, multiple answers)



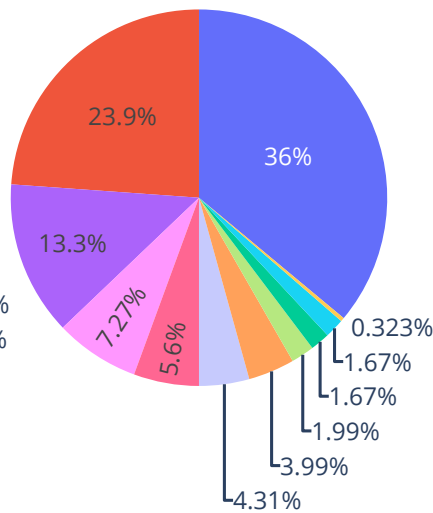
What tools and workflows do you use to debug Rust programs on which operating systems?

(total responses = 2109)

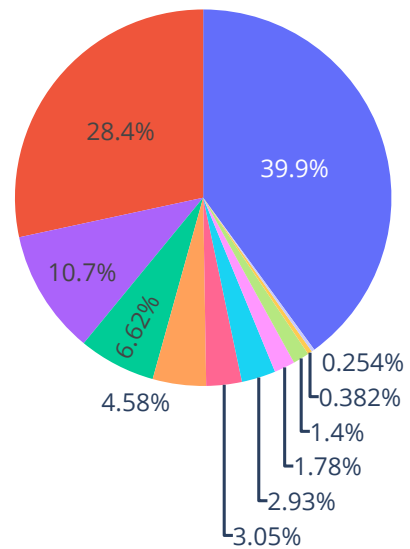
Linux



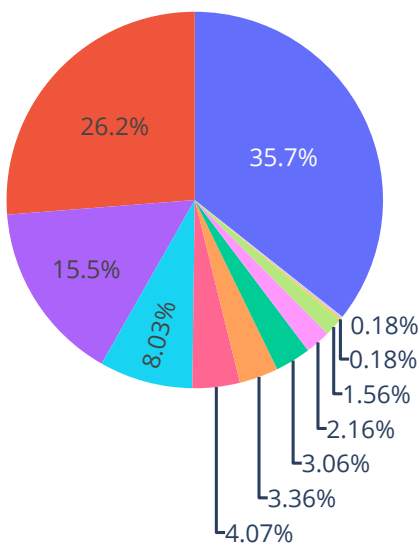
Windows



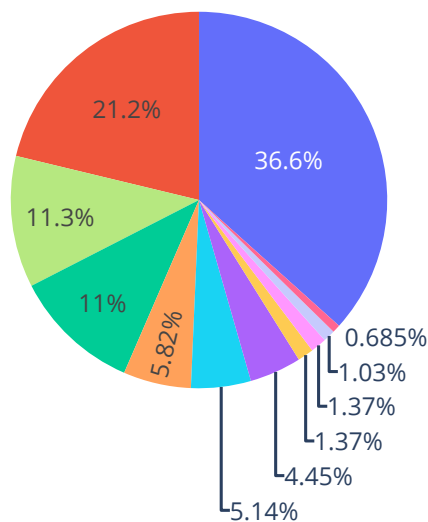
WSL



macOS



Other

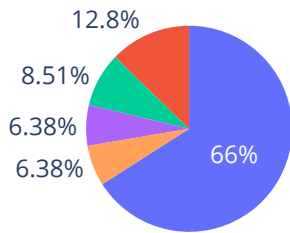


- Print debugging (e.g. println!)
- gdb (CLI)
- gdb (in IDE)
- I don't know, I just hit "Debug" in my IDE
- Visual Studio debugger
- WinDbg
- The dbg! macro
- lldb (in IDE)
- lldb (CLI)
- Special embedded debugger
- BugStalker

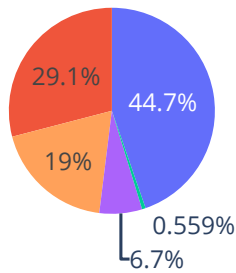
What tools and workflows do you use to debug Rust programs on which operating systems?

(total responses = 2109)

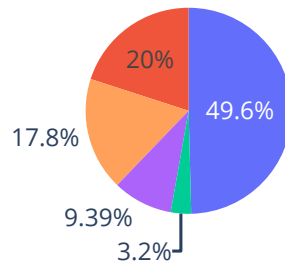
BugStalker



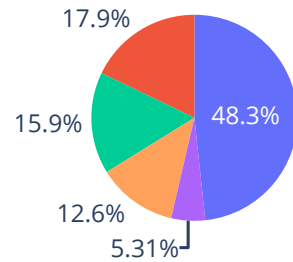
I don't know



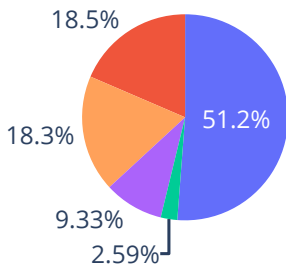
Print debugging



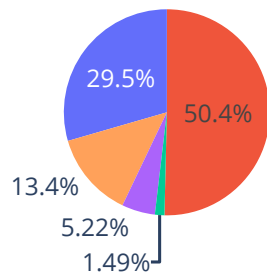
Special embedded debugger



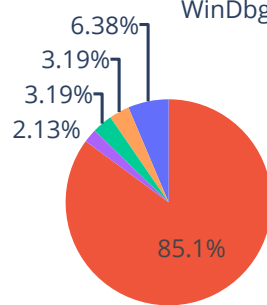
The dbg! macro



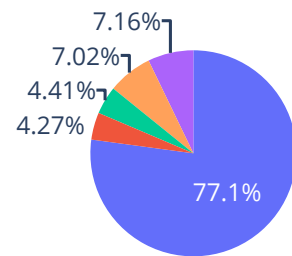
Visual Studio debugger



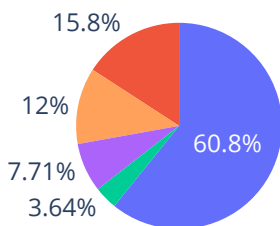
WinDbg



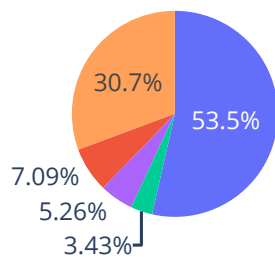
gdb (CLI)



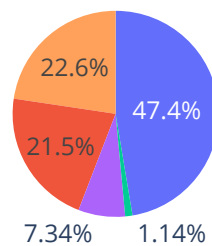
gdb (in IDE)



lldb (CLI)

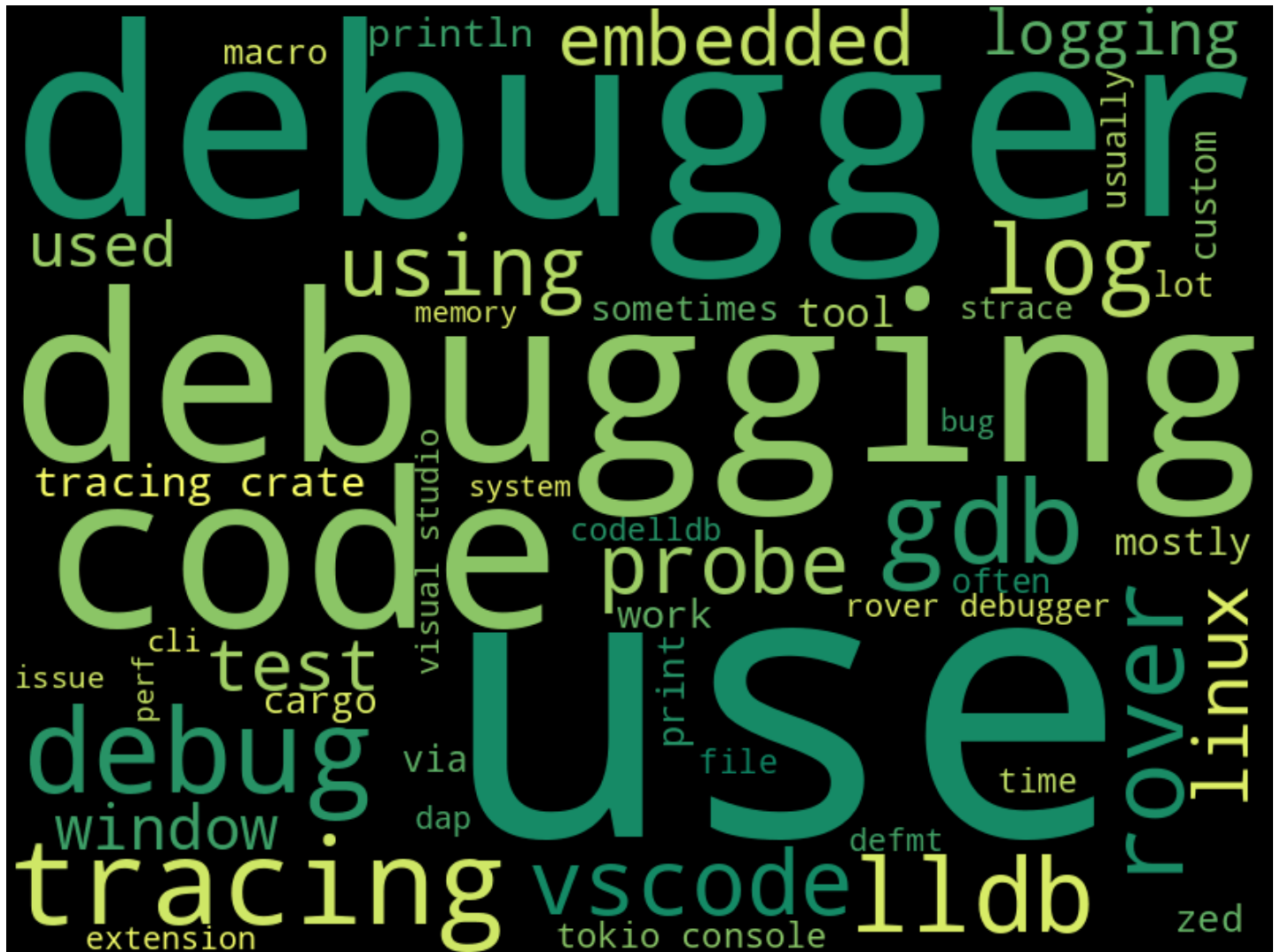


lldb (in IDE)



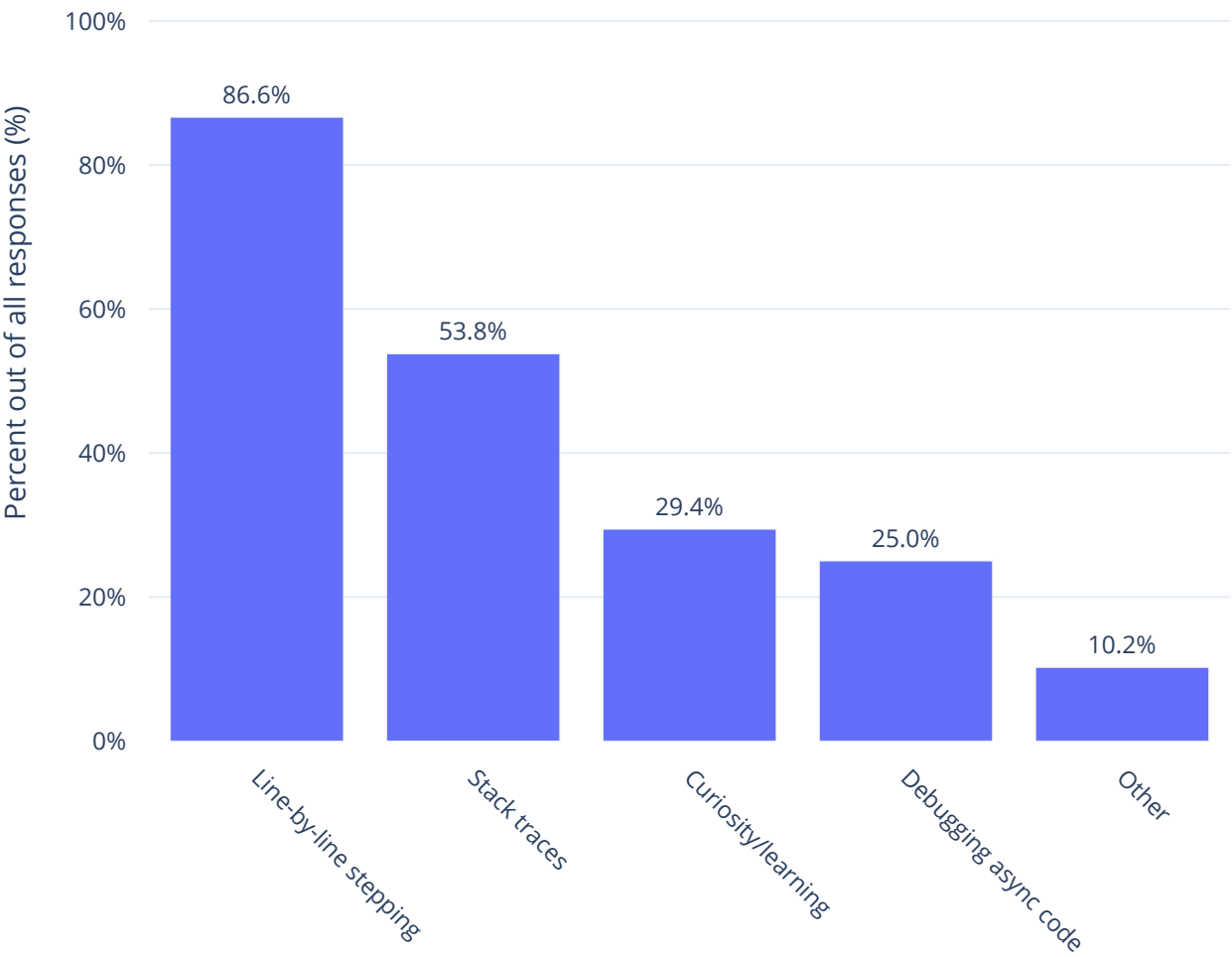
Linux Windows Other WSL macOS

Wordcloud of open answers for the previous chart:



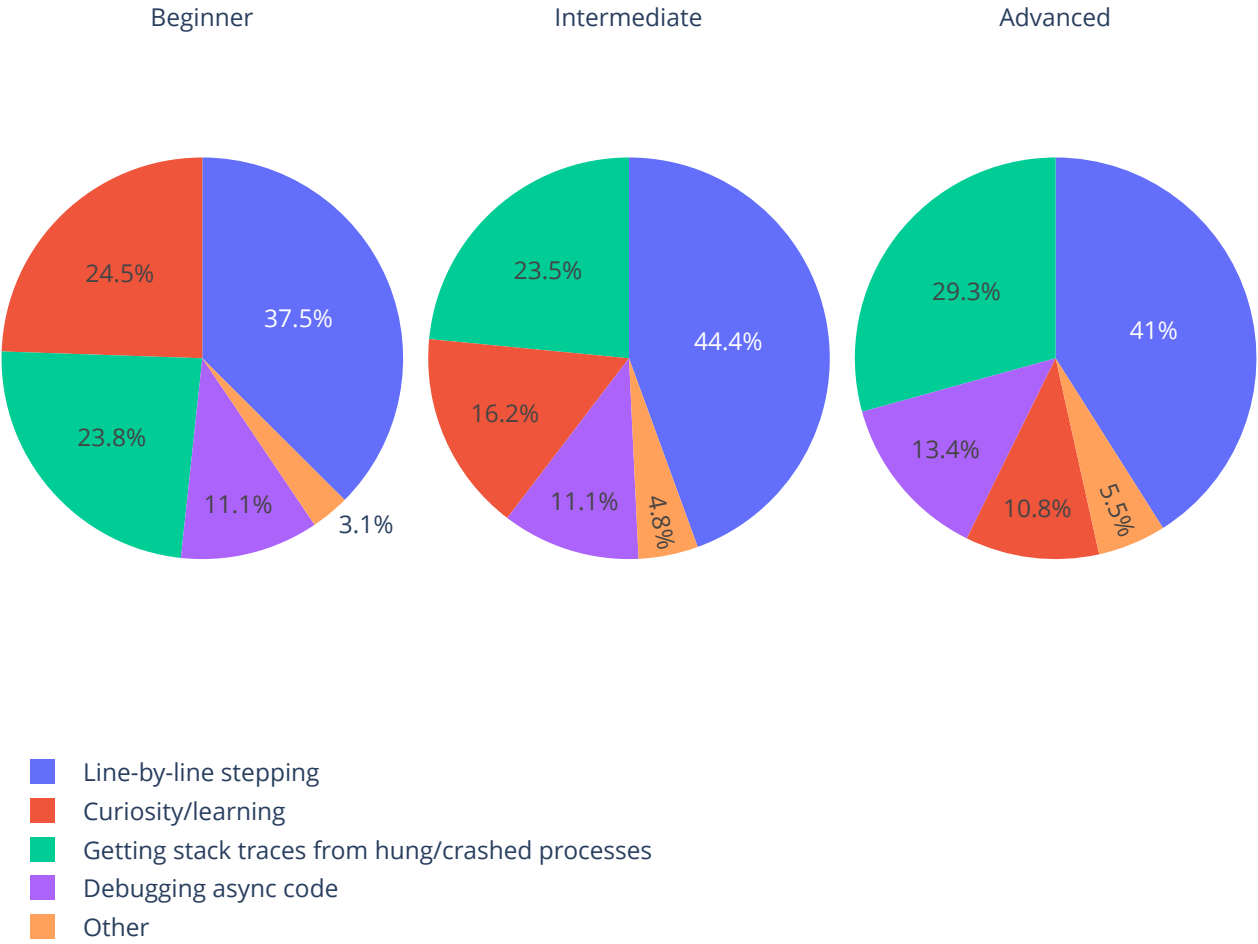
What are you using debuggers for?

(total responses = 1860, multiple answers)



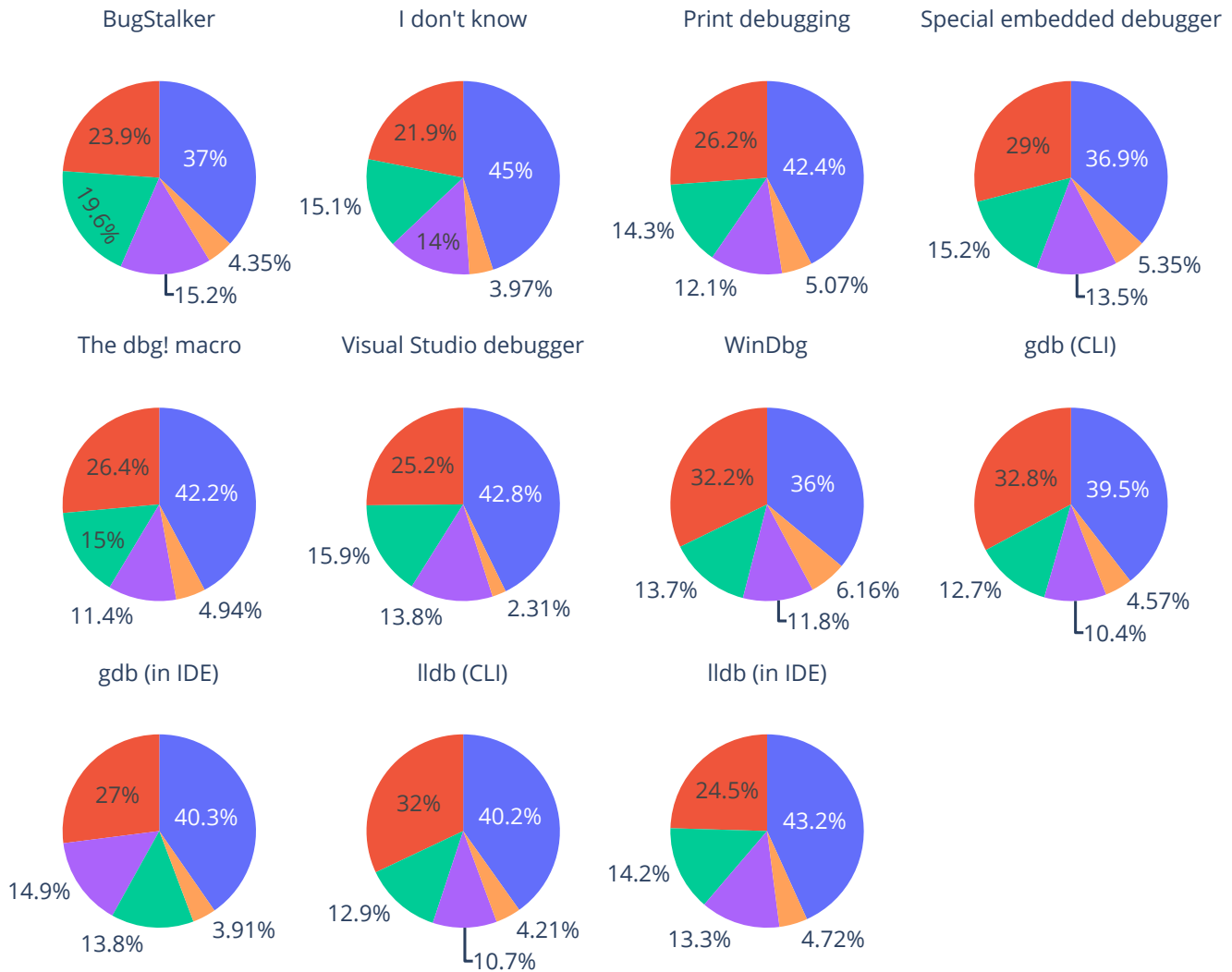
What are you using debuggers for? (based on expertise)

(total responses = 1860)



What are you using debuggers for? (based on debugger)

(total responses = 1860)

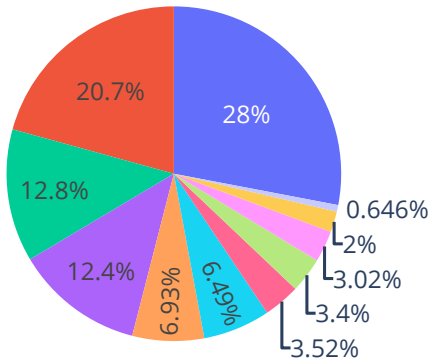


- Line-by-line stepping
- Getting stack traces from hung/crashed processes
- Curiosity/learning
- Debugging async code
- Other

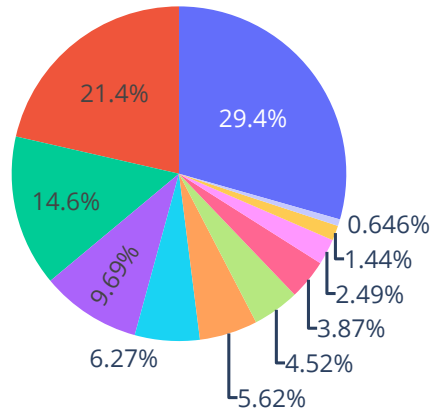
What are you using debuggers for? (based on debugger)

(total responses = 1860)

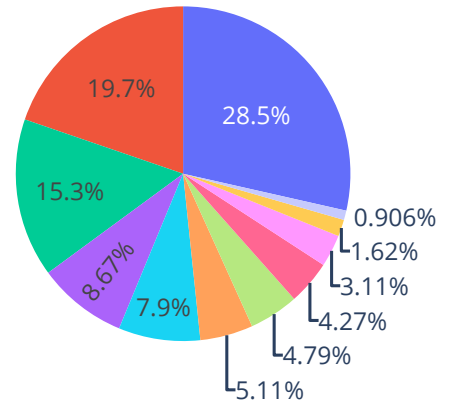
Stack traces



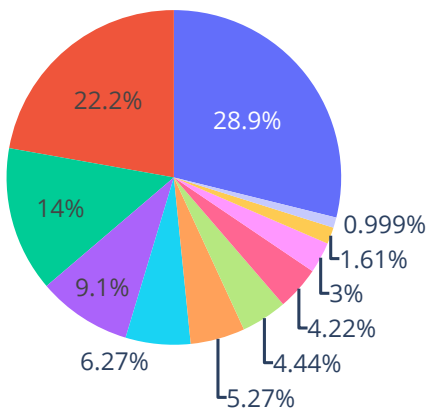
Line-by-line stepping



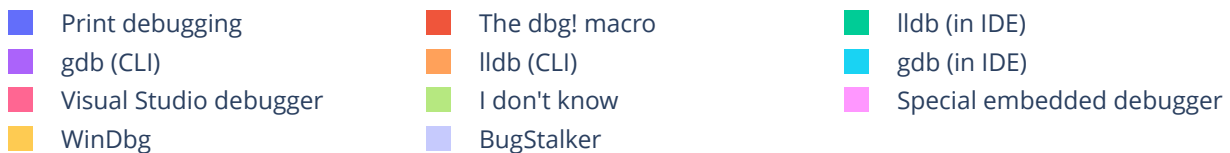
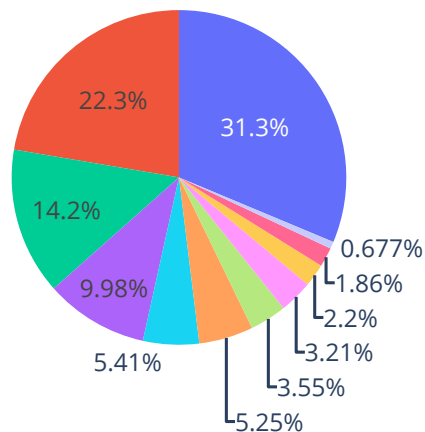
Debugging async code



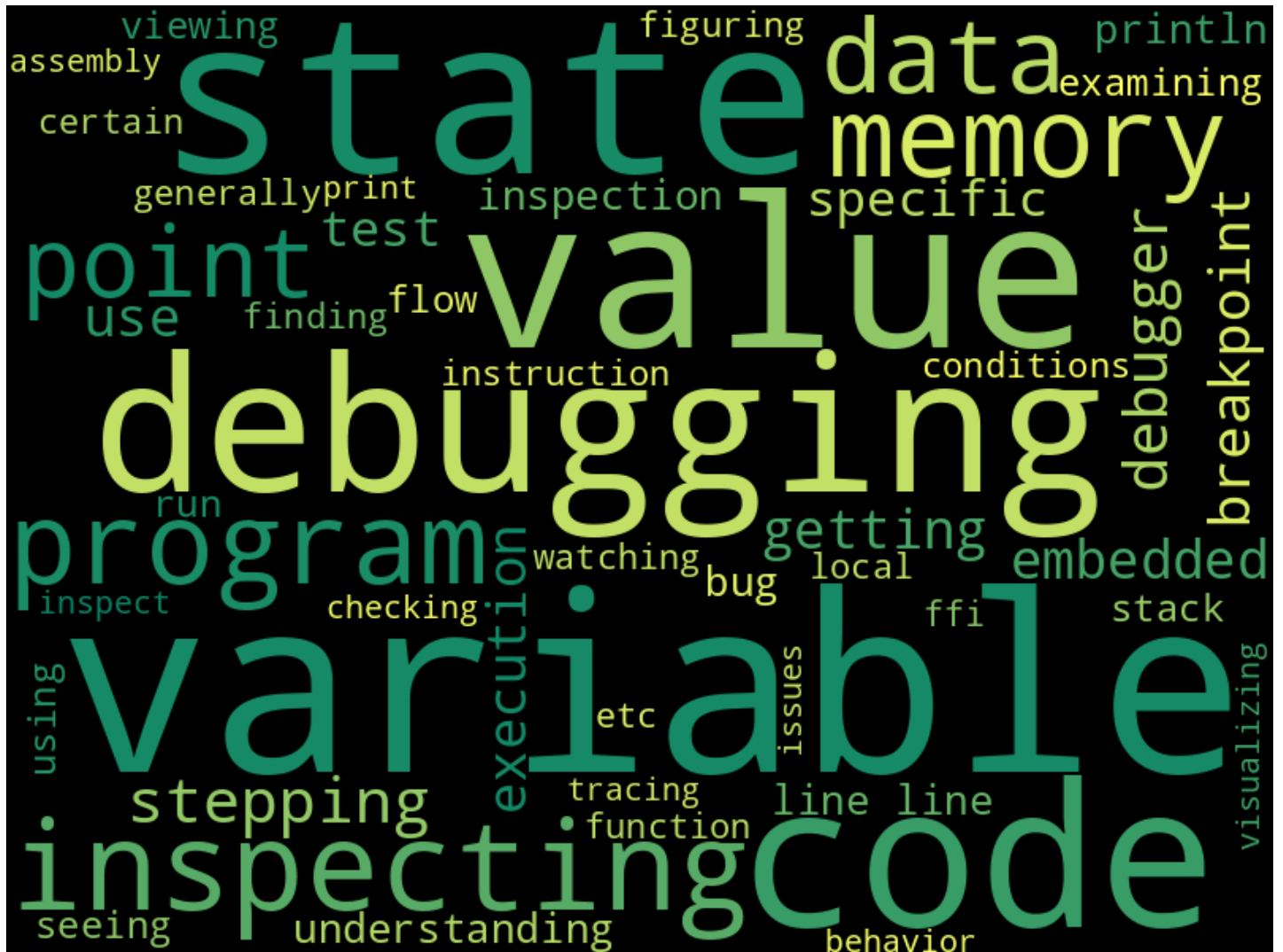
Curiosity/learning



Other

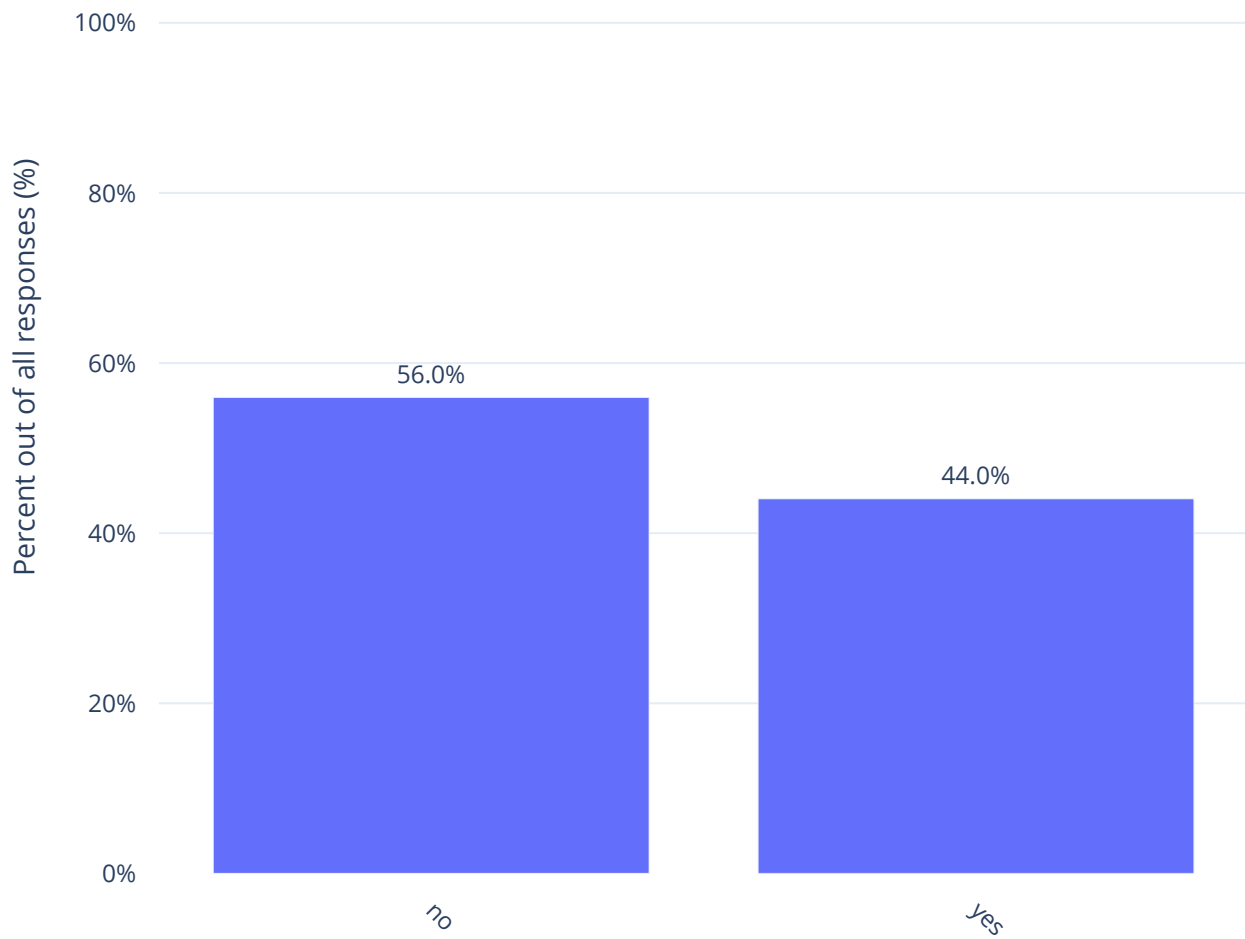


Wordcloud of open answers for the previous chart:



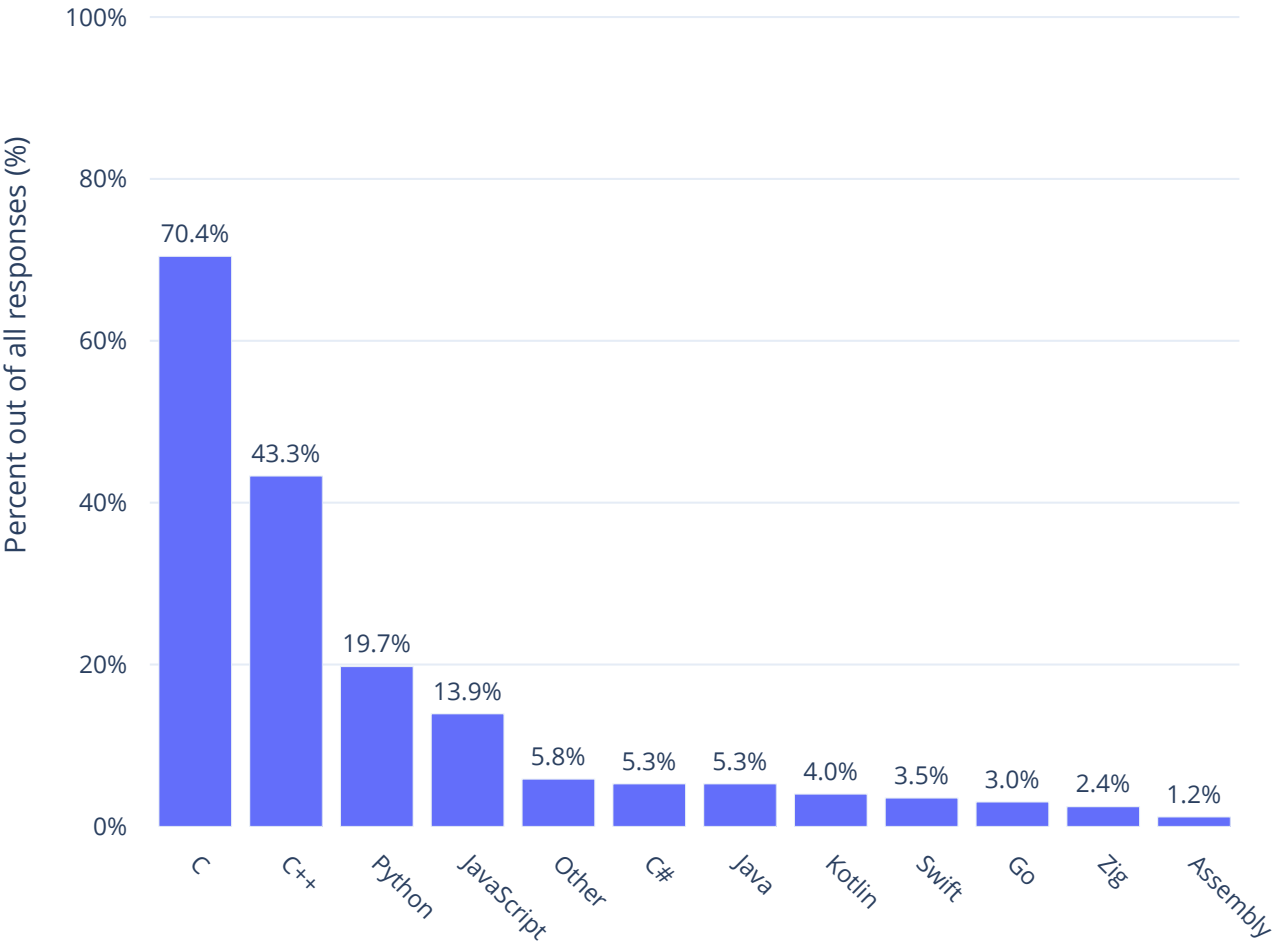
Do you debug programs that combine Rust with other languages?

(total responses = 2334, single answer)

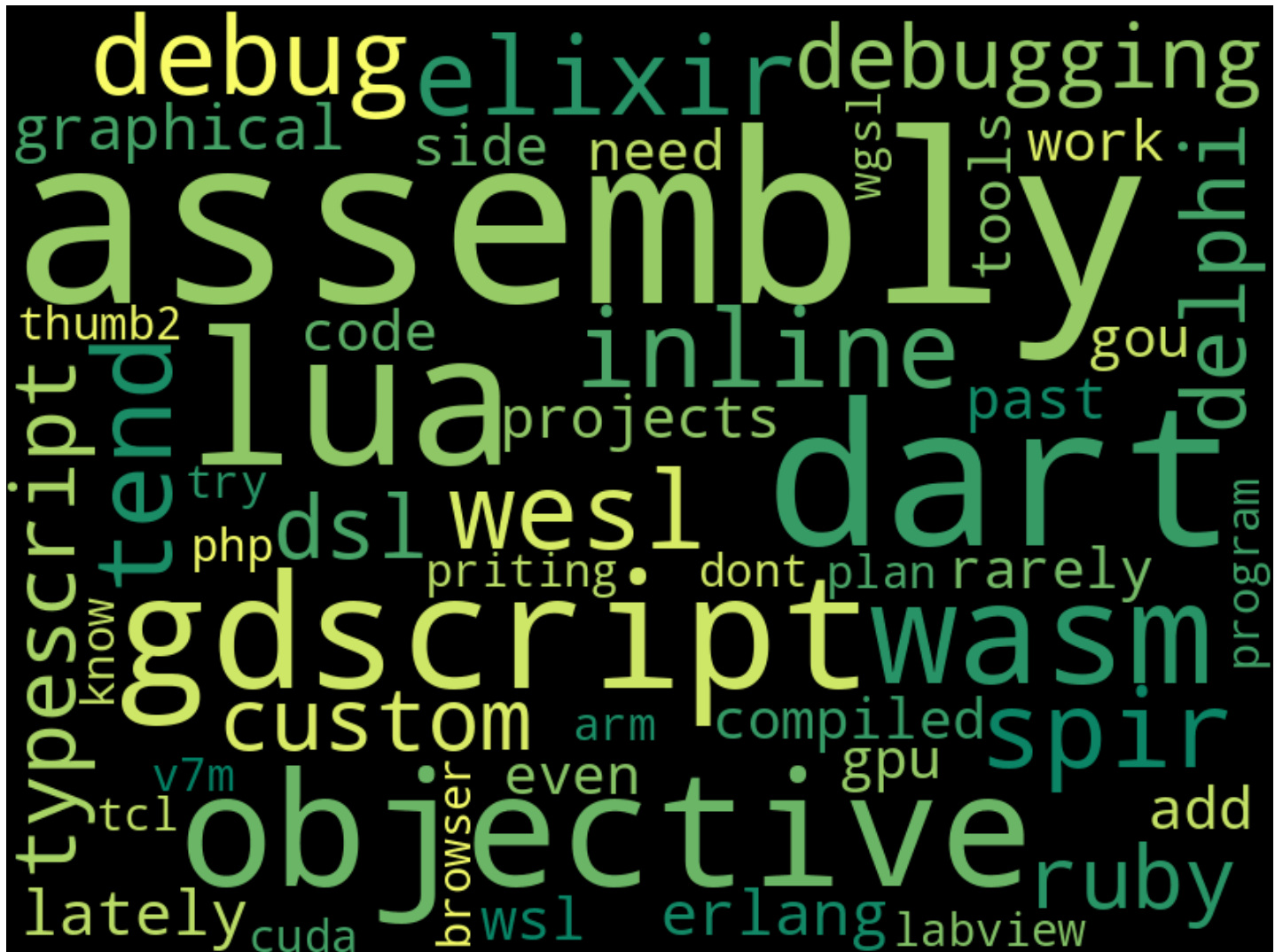


If you do, with which languages?

(total responses = 1028, multiple answers)

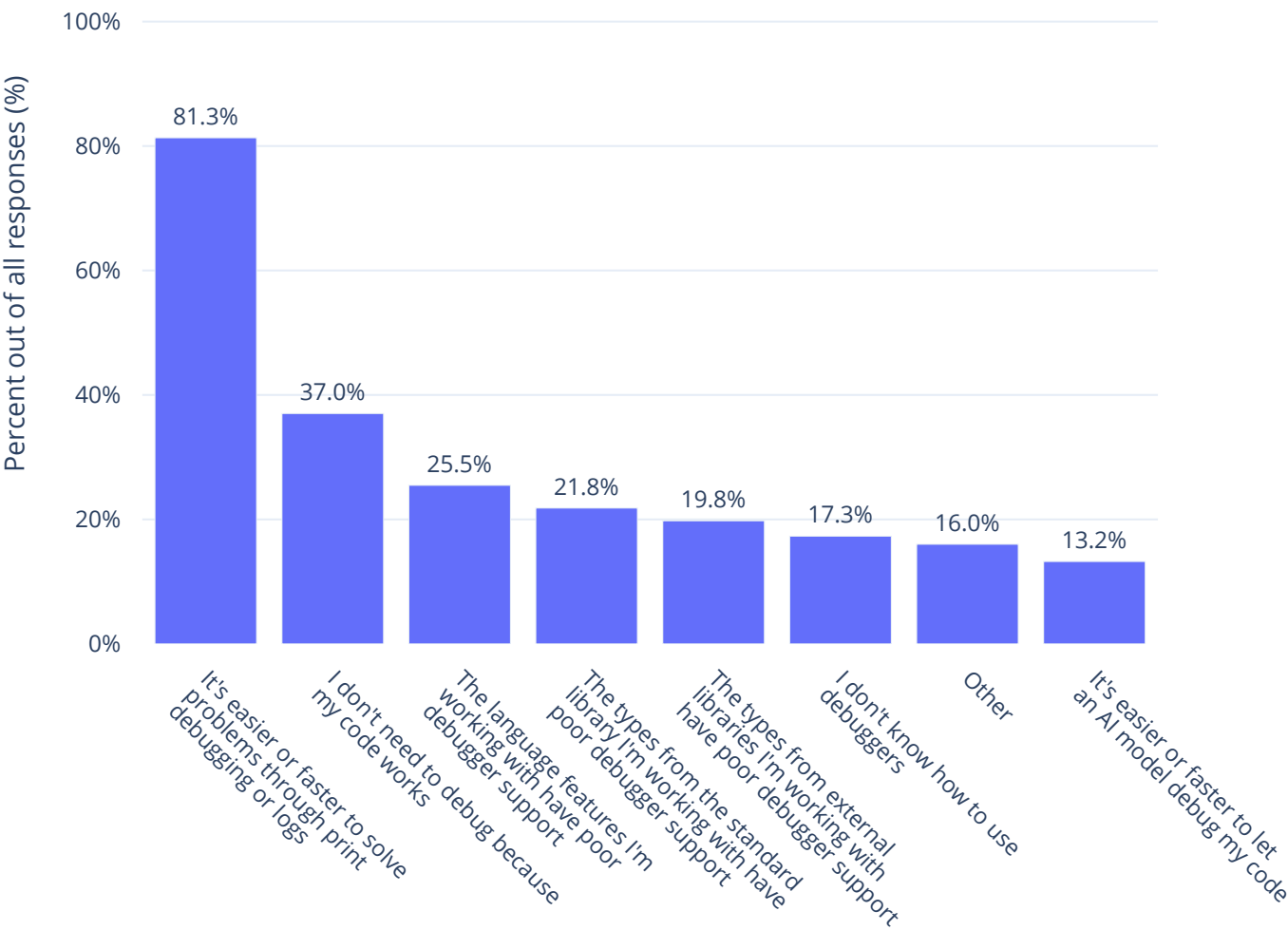


Wordcloud of open answers for the previous chart:

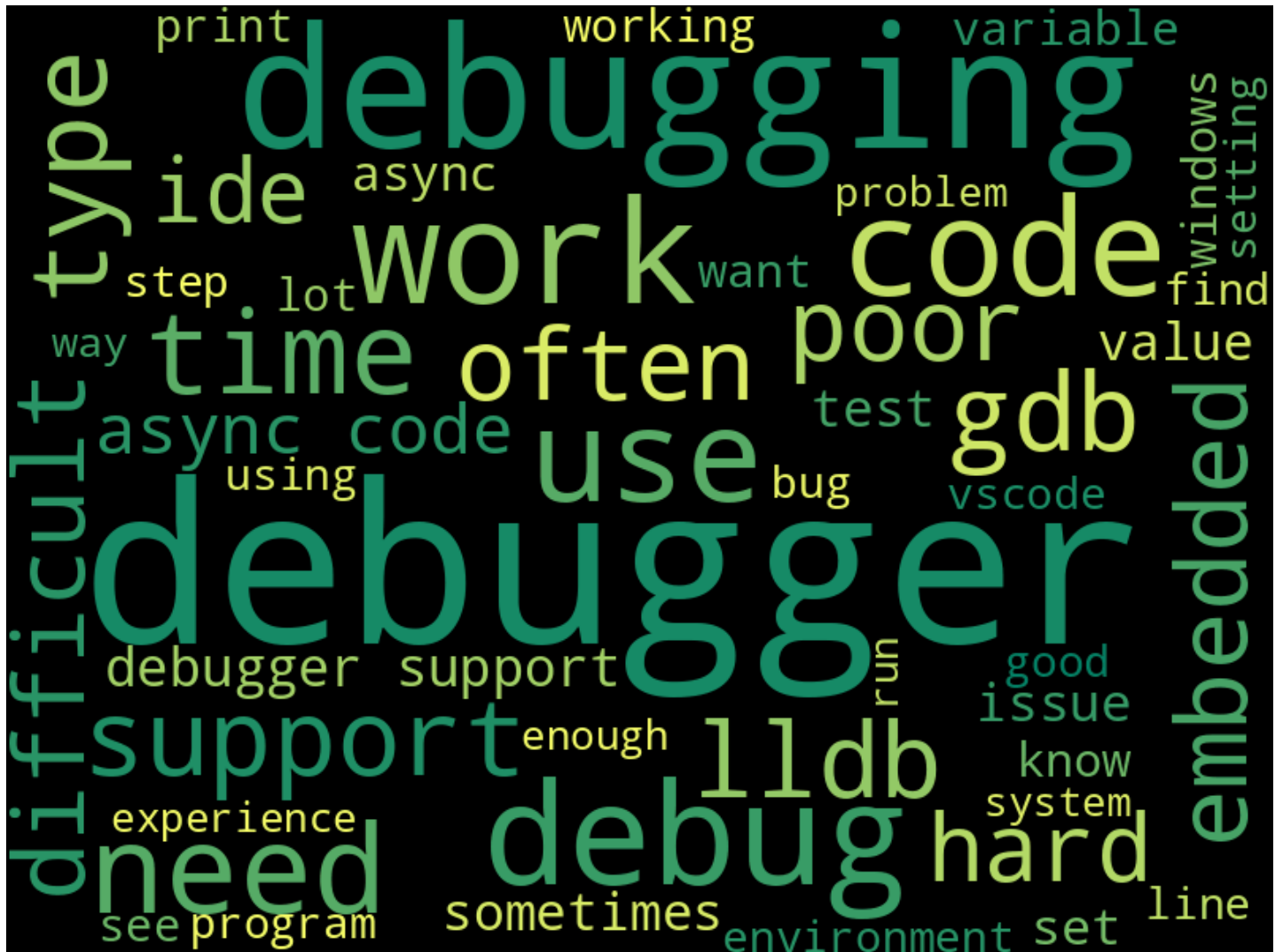


When you DON'T use a debugger, why don't you?

(total responses = 1943, multiple answers)



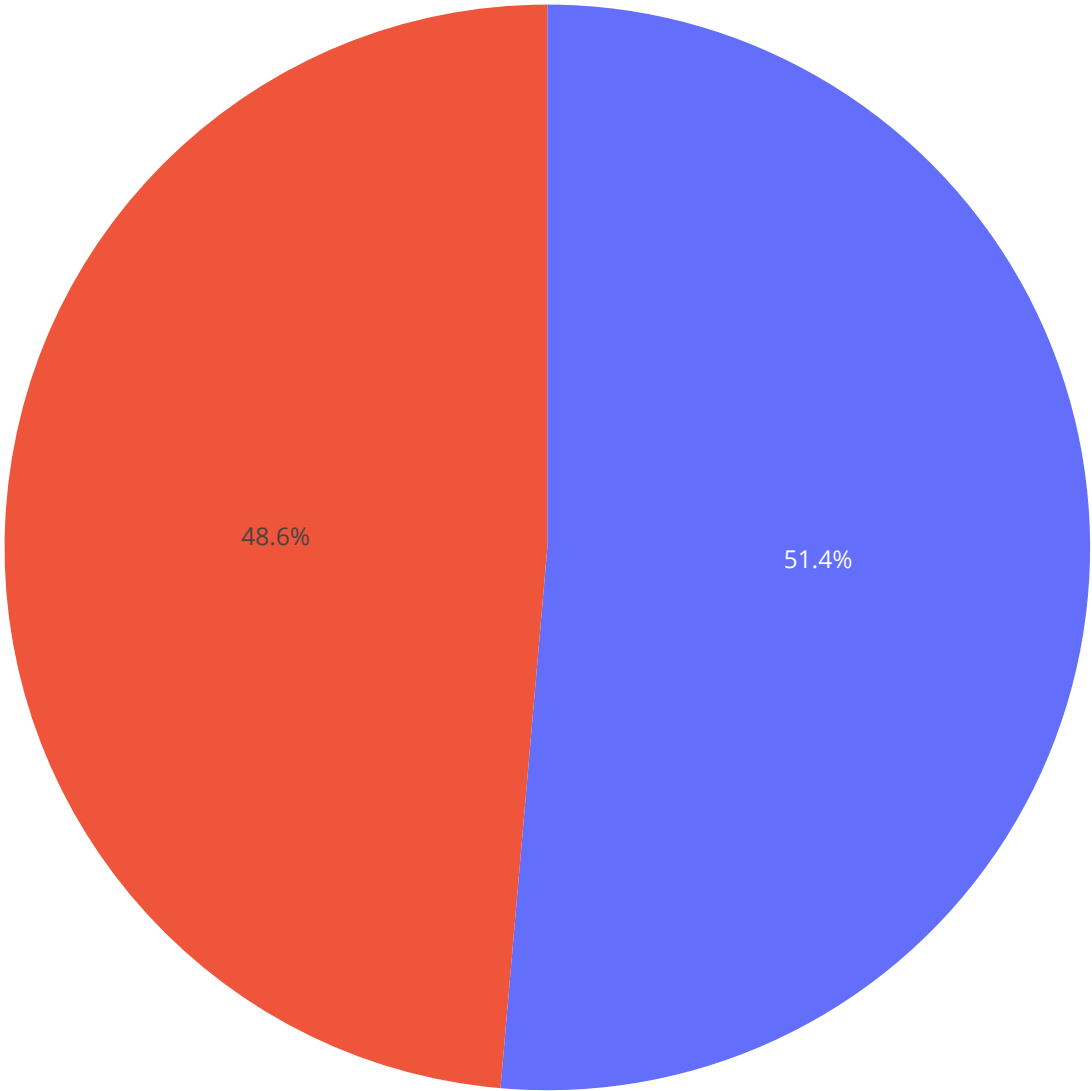
Wordcloud of open answers for the previous chart:



Do you experience any issues when trying to step through code with your debugger?

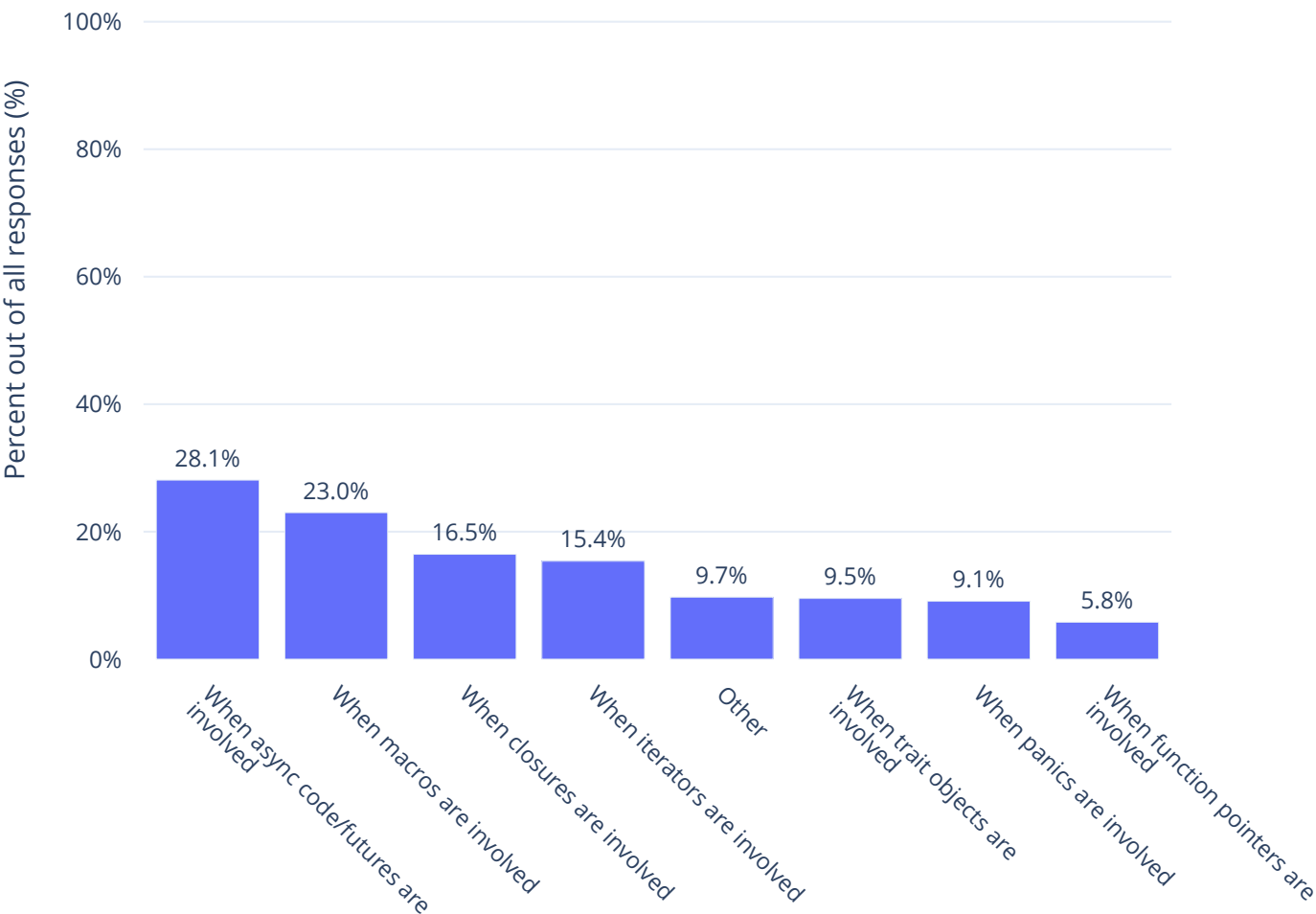
(total responses = 2036)

- Yes
- No

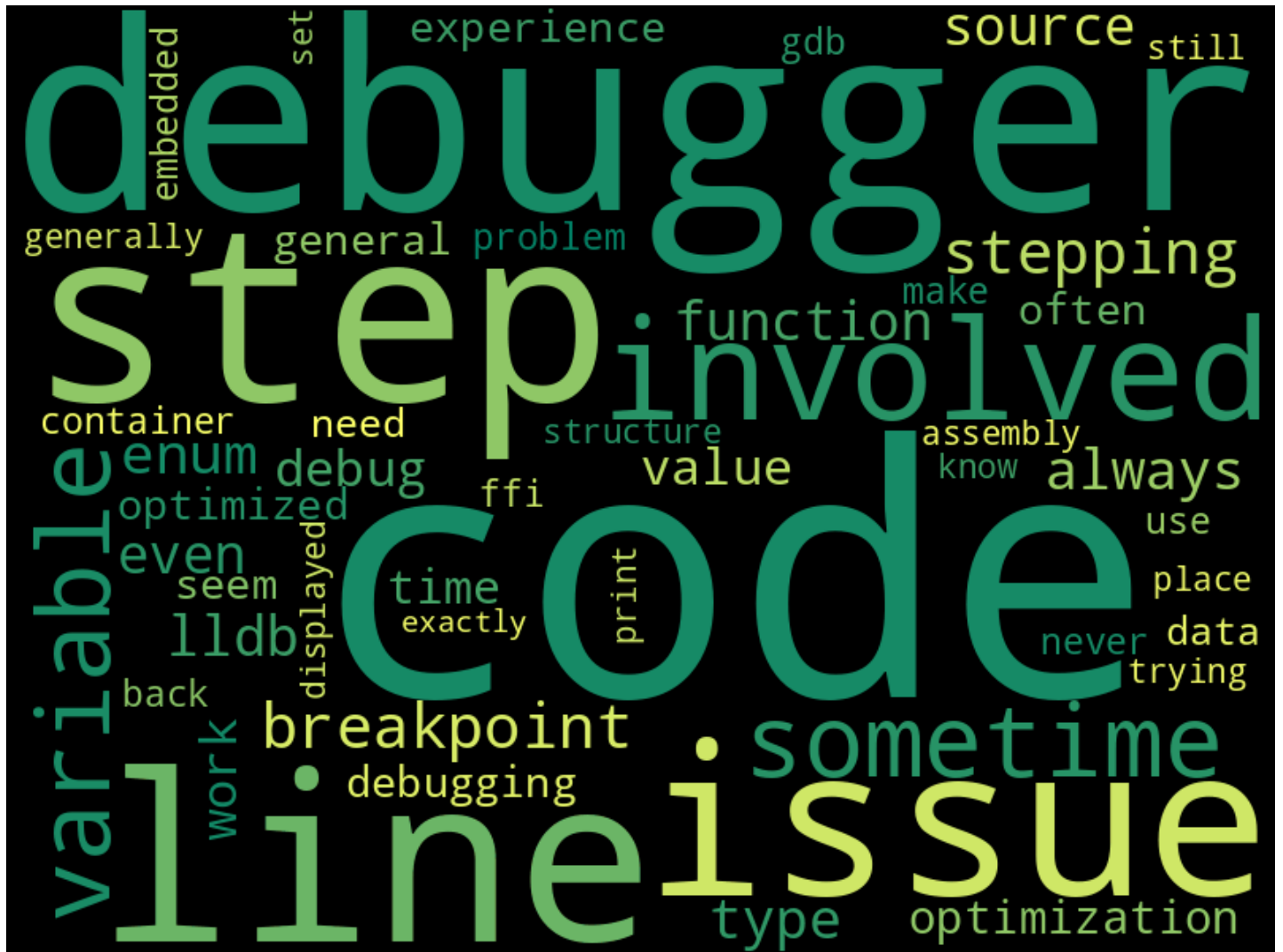


When do you experience issues with trying to step through code with your debugger?

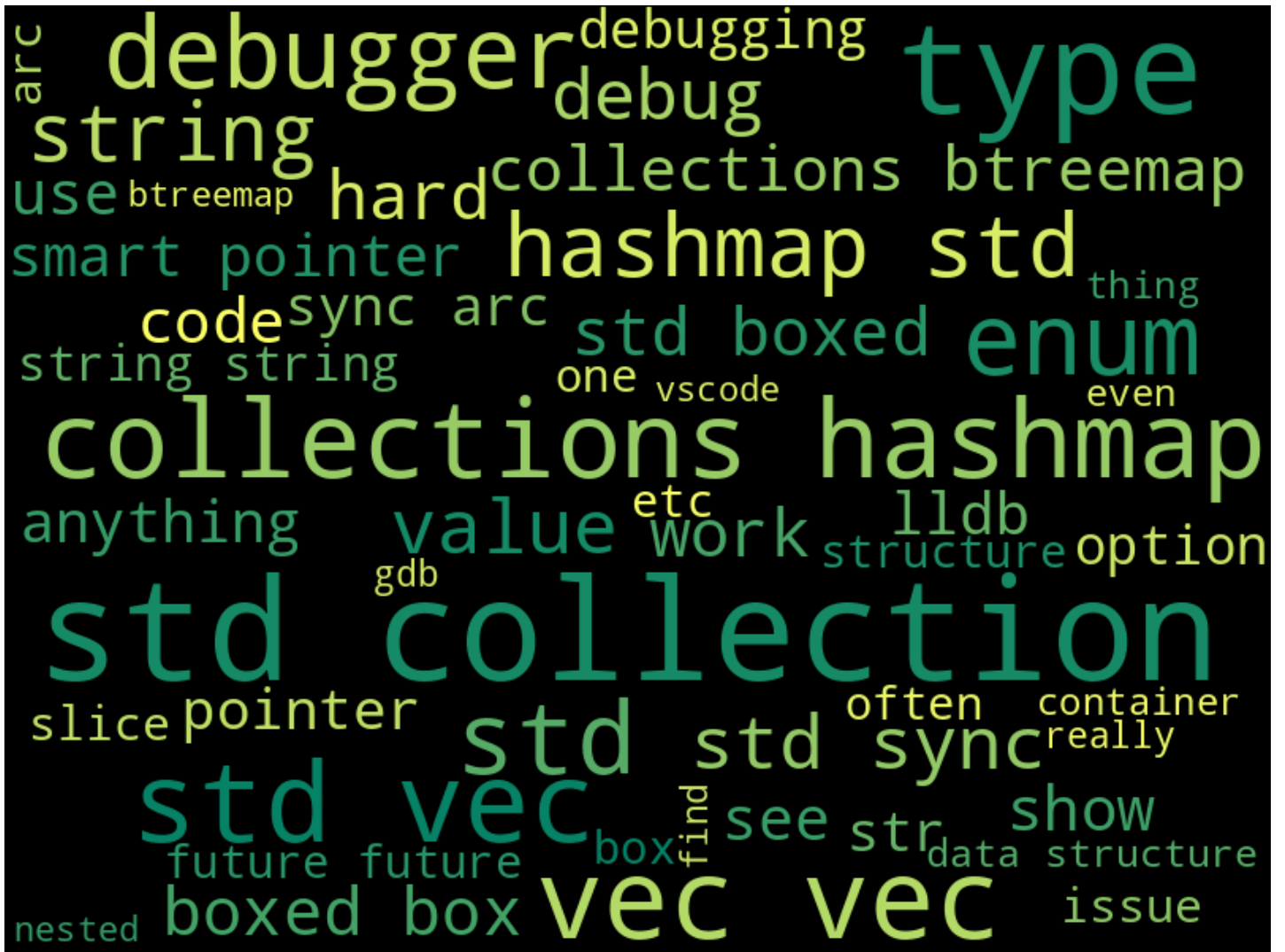
(total responses = 2036, multiple answers)



Wordcloud of open answers for the previous chart:



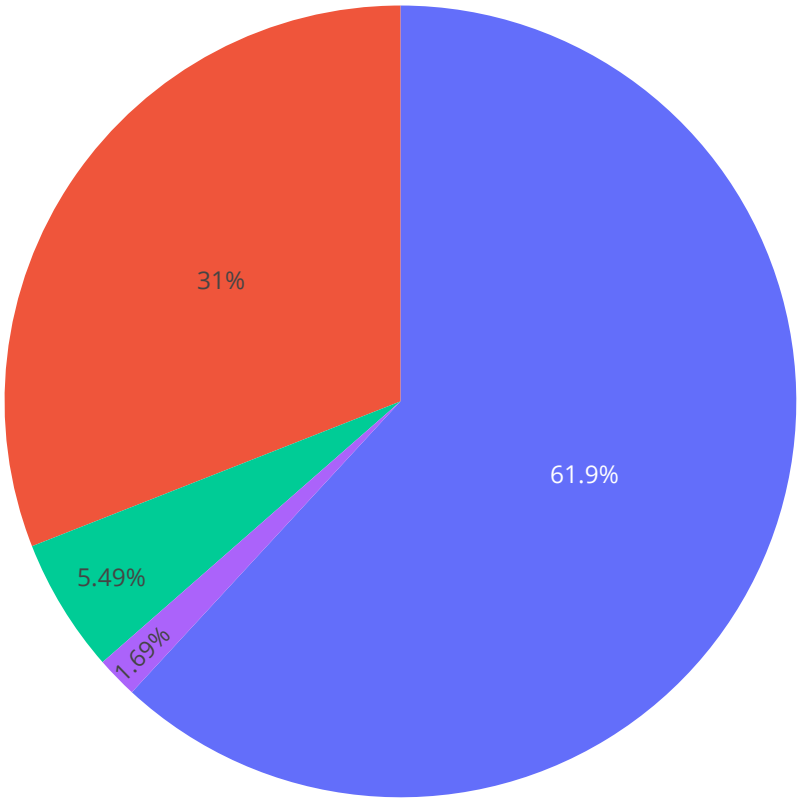
Wordcloud of open answers for the previous chart:



If you are a library author, are you aware of and using the debugger visualizer attribute?

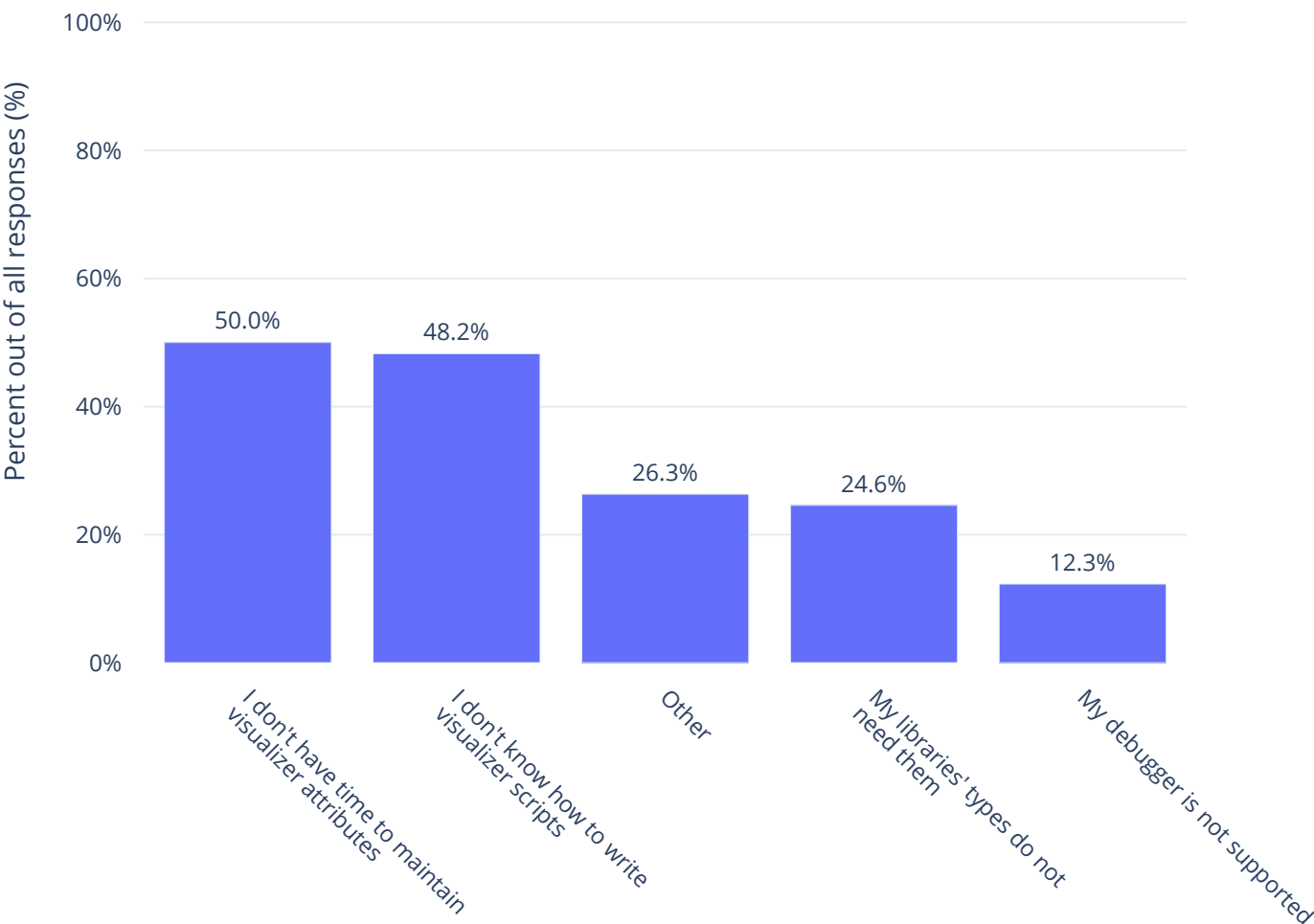
(total responses = 1896)

- I was not aware!
- I am not a library author
- I was aware, and do not use it
- I was aware, and already use it

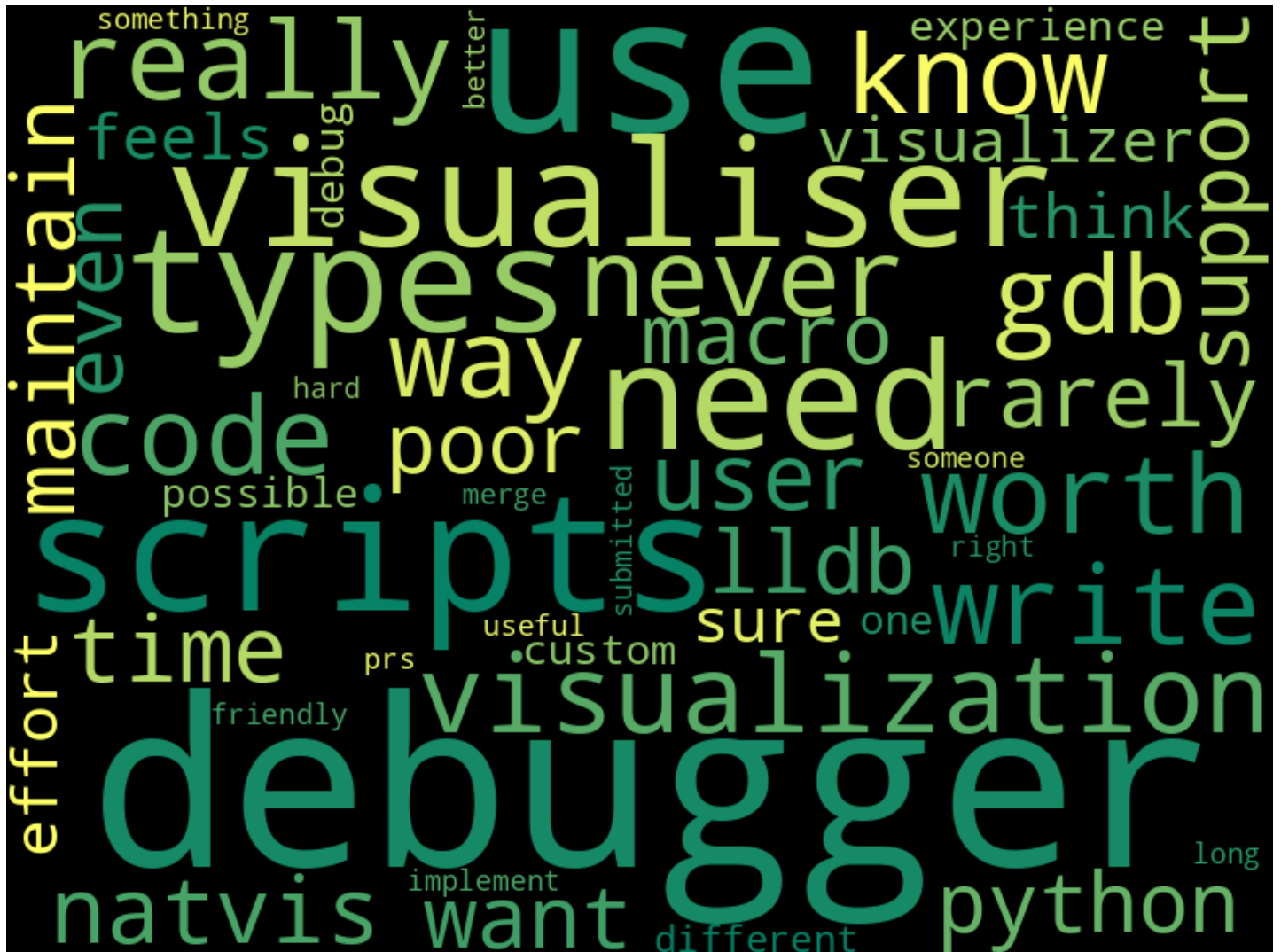


Why don't you use the debugger visualizer attribute?

(total responses = 114, multiple answers)

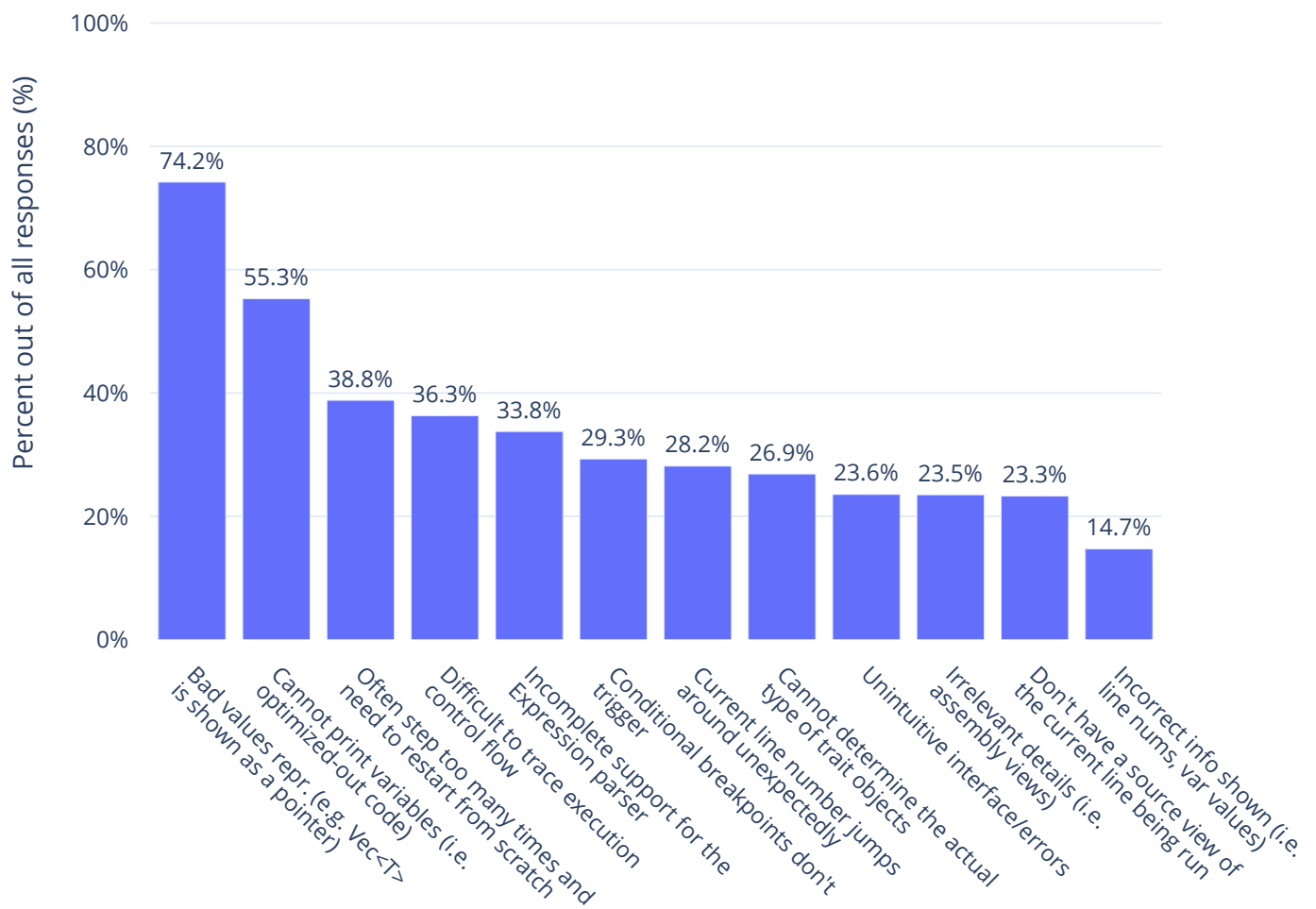


Wordcloud of open answers for the previous chart:



Which of these pain points have you experienced using a debugger with Rust?

(total responses = 1481, multiple answers)



Wordcloud of open answers for the previous chart:

